Camera-Incremental Person Re-Identification via Geometric Representation Evolution

Mang Ye, *Senior Member, IEEE*, Shipeng Zhang, Wuxuan Shi, Zhijie Lu, He Li, Xingyu Gao, *Member, IEEE*

***Abstract*—Person re-identification (ReID) is a core technology in large-scale intelligent surveillance systems. In practical deployments, surveillance networks evolve continuously as new cameras are incrementally integrated. As the network expands, the system not only encounters new identities but also captures previously known identities from novel camera views. However, existing LReID studies typically assume disjoint identity spaces across stages, failing to model the cumulative evolution of features induced by such expansion. To bridge this gap, we introduce Camera-Incremental Person Re-Identification (CIReID), a challenging setting that requires continuously updating representations from newly added cameras without access to historical raw data. Unlike traditional closed-set forgetting, the primary challenge in CI-ReID lies in feature space distortion and representation fragmentation across domain shifts. To address this, we propose Camera-Incremental Geometric Representation Evolution (CIGRE). First, we introduce a Dynamic Geometric Alignment (DGA) mechanism, which flexibly expands an Equiangular Tight Frame (ETF) classifier to provide a stable and scalable discriminative geometric skeleton for the evergrowing identity set. Building upon this stable structure, we propose a Cross-Camera Identity Consolidation (CCIC) strategy. By maintaining momentum-updated global prototypes, CCIC progressively aggregates the fragmented identity features revealed under novel views. Furthermore, we establish a new CI-ReID benchmark derived from standard datasets. Extensive experiments demonstrate that CIGRE effectively scales with longterm camera expansion and achieves state-of-the-art performance by strictly maintaining cross-camera consistency. Our code and models will be released.**

***Index Terms*—Lifelong learning, person re-identification, equiangular tight frame.**

I. Introduction

PERSON re-identification [2], [3], [4], [5] aims to match the same individual across spatio-temporal scenes, serving as a fundamental component of large-scale intelligent surveillance and urban sensing systems. In practical deployments, surveillance networks are not always static. Instead, they evolve continuously as new cameras are incrementally integrated over time. This dynamic network expansion necessitates LReID models capable of adapting to newly streamed data [6], [7], [8], [9]. However, a critical hurdle emerges during the expansion process: newly deployed cameras inevitably capture a highly overlapped mixture of completely unseen identities and previously registered individuals appearing under entirely novel viewpoints [10], [11], [12]. Due to strict privacy regulations and storage constraints, retaining massive historical raw data is typically prohibitive. Therefore, LReID models are required to continuously incorporate new appearance cues to instantly update existing identity representations [13], [14], [15], [16]. Effectively consolidating these evolving identities across growing camera networks, without revisiting past data, remains a formidable challenge in scalable ReID applications.

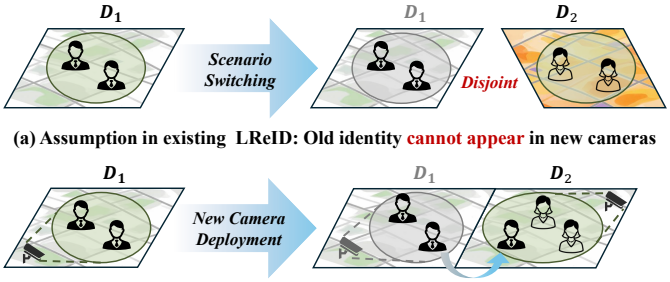


Fig. 1. Comparison between conventional lifelong person re-identification (LReID) and the proposed camera-incremental person re-identification (CI-ReID). Traditional LReID assumes disjoint identity spaces across stages under scenario switching, whereas CI-ReID models the realistic expansion of surveillance networks, where newly deployed cameras introduce both previously seen identities under novel views and unseen identities, resulting in overlapping identity spaces across stages.

Recent advancements in LReID have made notable progress in mitigating catastrophic forgetting across sequentially arriving domains [17], [18], [19], [20], [21]. However, these prevailing paradigms typically operate under a restrictive assumption: they presume completely disjoint identity spaces across different learning stages. When applied to real-world camera scenarios with overlapping identity spaces, this assumption fundamentally breaks down. Specifically, when a previously known identity reappears under a novel camera with significant domain shifts (e.g., varying lighting or viewpoints), existing disjoint-focused methods fail to model the cumulative feature evolution. They either rigidly constrain historical knowledge or mistakenly isolate new perspectives as independent entities

This work was supported by the Brain Science and Brain-like Intelligence Technology—National Science and Technology Major Project under Grant 2022ZD0208700, and the National Natural Science Foundation of China under Grants 62376264 and 62506269. The numerical calculations in this paper were supported by the supercomputing system at the Supercomputing Center of Wuhan University.

M. Ye, S. Zhang, W. Shi, Z. Lu and H. Li are with the School of Computer Science, the National Engineering Research Center for Multimedia Software, Wuhan University, Wuhan, China. E-mail:{yemang, shipengzhang, wuxuanshi, zhijielu, lihe404}@whu.edu.cn

X. Gao is with the Institute of Microelectronics, Chinese Academy of Sciences, Beijing 100029, China, and also with the University of Chinese Academy of Sciences, Beijing 100049, China E-mail:{gxy9910@gmail.com}

Corresponding author: *Xingyu Gao, He Li*.

A preliminary version of this work has appeared in IJCAI 2025 [1].

0000–0000/00$00.00 © 2021 IEEE

[22], [23], [24], [25]. Consequently, the representation of a single identity becomes shattered across different camera domains, leading to severe representation fragmentation. Simultaneously, forcing disjoint learning objectives onto an overlapping and continuously expanding identity set inevitably disrupts the global discriminative topology, resulting in feature space distortion.

To bridge the gap between lifelong learning assumptions and real-world surveillance requirements, we formalize a novel and highly challenging scenario: Camera-Incremental Person Re-Identification, as shown in the Fig. 1. Unlike the traditional domain-incremental paradigm where data streams do not overlap, CI-ReID faithfully recreates the continuous expansion process of a camera network. In this setting, the system incrementally receives data solely from newly deployed cameras at each stage. Crucially, these incoming data streams contain a highly overlapped mixture of completely unseen identities and previously encountered individuals appearing under novel camera viewpoints. By formalizing this scenario, we shift our research focus from mitigating traditional closed-set forgetting to explicitly addressing representation fragmentation and feature space distortion caused by cross-camera domain shifts, as shown in the Fig. 2.

CI-ReID involves two conceptually distinct problems. The first is cross-stage identity correspondence acquisition, which determines whether an observation from a newly deployed camera belongs to a previously seen identity. The second is cross-camera representation evolution, which updates the representation of a matched identity using its newly observed appearance while preserving the global embedding structure. This work focuses on the second problem. Our primary benchmark adopts the standard supervised ReID annotation protocol, in which identity labels are globally consistent across cameras and stages. Meanwhile, the replay-free constraint remains strict: after a stage is completed, its raw images cannot be revisited. Cross-stage correspondence may in principle be supplied by manual annotation, retrieval, clustering, or pseudo-label association. Once correspondence is given or estimated, however, the model must still absorb new camera-specific evidence without fragmenting the identity representation or distorting the previously established feature geometry.

To address cross-camera representation evolution under this replay-free constraint, we propose Camera-Incremental Geometric Representation Evolution. CIGRE separates global geometric stability from identity-level visual consolidation. At the global level, Dynamic Geometric Alignment maintains a set of fixed class-level targets. Previously established targets remain unchanged, while highly separated targets are appended for newly observed identities. This construction prevents the classifier geometry from being repeatedly pulled toward the most recent camera domain. At the identity level, Cross-Camera Identity Consolidation maintains data-driven visual prototypes. A prototype is initialized from the normalized mean representation available when an identity is first observed and is subsequently consolidated with observations from newly introduced cameras. Thus, the geometric targets and identity prototypes have complementary roles: the former provide a stable global reference, whereas the latter accumulate evolving visual evidence.

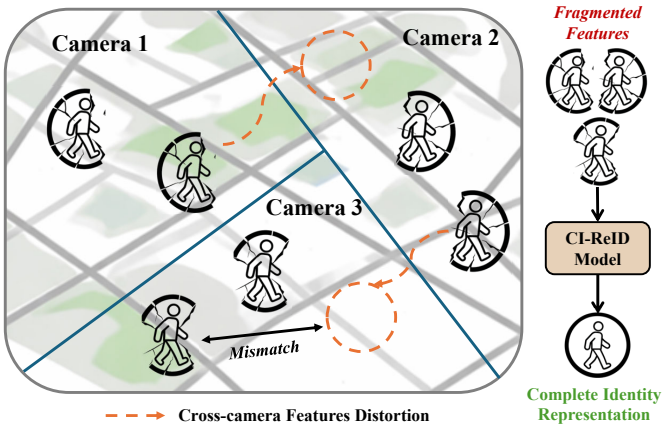


Fig. 2. Illustration of the core challenge in CI-ReID. As the camera network expands, the same identity may reappear under newly introduced cameras with significant cross-camera domain shifts, causing feature distortion and representation fragmentation. Such fragmented observations can easily lead to mismatches unless they are progressively consolidated into a complete identity representation.

The geometry used by DGA is inspired by simplex equiangular tight frames [26], [27], [28] and Neural Collapse, but it should not be interpreted as an exact finite-class simplex-ETF. A finite-K-class simplex-ETF has pairwise inner product $-1/(K-1)$. In contrast, when a nonempty orthogonal complement is available, DGA appends a target orthogonal to the span of the established targets without changing them. DGA is therefore an ETF-inspired orthogonalized fixed-target approximation designed for a growing identity space.

This manuscript significantly expands upon our preliminary research on LReID [1], the conference version of which was published in IJCAI 2025. In this journal version, we contribute the following major improvements:

- We formalize supervised Camera-Incremental Person Re-Identification, where globally consistent identity annotations provide cross-stage correspondence, newly deployed cameras contain both repeated and unseen identities, and historical raw images are unavailable after each stage. This protocol isolates the problem of representation evolution under camera-network expansion.
- We introduce Dynamic Geometric Alignment, an ETF-inspired incremental fixed-target mechanism. DGA freezes established identity targets and appends highly separated targets for newly observed identities, thereby reducing global geometric distortion without historical replay.
- We propose Cross-Camera Identity Consolidation, which maintains data-driven identity prototypes through feature-based initialization and normalized stage-level momentum consolidation. CCIC progressively integrates complementary camera-specific appearances and reduces identity representation fragmentation.
- We establish camera-incremental protocols on Market-1501 and MSMT17 and conduct comprehensive evaluations across multiple camera orders. In addition to the main comparison and component ablations, we also

analyze identity overlap, initialization sensitivity, camera discrepancy, prototype evolution.

II. Related Work

A. Incremental Learning

Incremental learning [29], [30], [31] aims to integrate new knowledge into streaming data. This can be further subdivided into three main categories: task-incremental learning (TIL), class-incremental learning (CIL) [32], and domain-incremental learning (DIL) [33]. Specifically, TIL and CIL need to handle situations where new categories appear, requiring the retention of knowledge during the learning process [10]. DIL primarily addresses situations where the label space is fixed but the data involves new domains [34], [35]. For incremental learning, typical solutions include reproducing historical knowledge through data replay, and preserving existing knowledge using knowledge distillation or parameter regularization. Recent advancements also involve using bias correction to mitigate model bias, or expanding the network structure as the data evolves. Despite some progress, traditional CIL methods are typically built on the assumption of a closed-set class hierarchy, with the core objective of learning and maintaining stable discriminative boundaries for seen classes. However, this assumption differs significantly from the nature of ReID tasks. ReID focuses more on metric learning and retrieval in open-set scenarios, aiming to learn a discriminative embedding space characterized by the relative distances between features, rather than relying on fixed classification decision boundaries. Therefore, directly applying the rigid classification boundaries emphasized in CIL to ReID scenarios often limits the model's generalization to unknown identities during the inference.

B. Person Re-Identification

Person re-identification aims to match the same pedestrian across the same or different camera views [4], [36], [37], [38]. Early research relied primarily on manually designed features, utilizing low-level cues such as color, texture, and local descriptors to characterize pedestrian appearance. However, these methods rely on hand-constructed features, resulting in limited robustness and poor generalization in complex environments. With the emergence of large-scale labeled datasets and the rapid development of deep learning, ReID has gradually transitioned to a data-driven representation learning paradigm [39], [40], [41]. Convolutional neural networks and transformer-based models that have emerged in recent years, have significantly enhanced discriminative feature learning capabilities, leading to substantial performance improvements on standard benchmark datasets [36], [42], [43]. Traditional ReID research typically assumes that the camera network and identity space are fixed, and focused on optimizing feature separability under controlled settings [9], [44], [45]. Despite significant progress, such static assumptions often fail to hold in real-world deployments. Surveillance systems are constantly evolving with the introduction of new cameras, leading to dynamic changes in data distribution and identity space [46], [47], [48]. Models trained on static datasets struggle to adapt to these dynamic environments. This discrepancy highlights the limitations of traditional ReID methods, prompting a shift from static recognition to adaptive and evolutionary representation learning frameworks.

C. Lifelong Person Re-Identification

LReID has emerged as a key paradigm for addressing data evolution problem in the deployment of long-term ReID systems in recent years [49], [50], [15], [51], [52]. Unlike traditional static setups, LReID aims to enable models to continuously adapt to dynamic data streams while retaining previously acquired knowledge [53]. Existing LReID methods can be broadly categorized into three types: replay-based methods [18], [6], [17], which simulate historical data distributions by storing or generating historical samples; knowledge distillation-based methods [54], [55], [19], [56], [57], which preserve previously learned representations by restricting model updates; and domain style transformation-based methods [58], [59], [60], which assign domain offsets based on feature distributions under different camera environments. These methods have been shown to be effective in mitigating catastrophic forgetting. Despite progress, most existing LReID methods still inherit the basic framework of traditional incremental learning. They typically assume that the sets of identities at different stages are disjoint and that each identity feature can be fully learned upon its first appearance [61], [62], [63]. However, in real-world camera networks, the same identity may appear repeatedly at different camera views across multiple stages, and its visual representation is gradually enriched rather than presented completely at once. This gap causes existing LReID methods to treat new observations of the same identity as independent instances, failing to effectively integrate cross-camera information. Therefore, they struggle to capture the identity evolution and handle the representation fragmentation in camera incremental settings.

D. Neural Collapse and ETF Geometry

Neural Collapse [64], [65], [66] describes a terminal-training phenomenon in which within-class features contract, centered class means align with their classifier directions, and different classes approach a simplex-ETF configuration [67], [68], [69]. This observation has motivated fixed-classifier and ETF-guided representation learning methods, which use predefined class directions to promote balanced inter-class separation. Most existing formulations, however, assume a stationary and fixed class set. CI-ReID violates this assumption because identities are added over time, previously observed identities acquire new camera-specific appearances, and historical samples cannot be jointly revisited. Consequently, a learnable classifier may drift toward the most recent camera, whereas recomputing an exact simplex-ETF after every expansion would alter established class directions. DGA instead converts the geometric prior into an incremental fixed-target mechanism that preserves old directions and approximately maintains a highly separated target structure as the identity set grows.

III. METHOD

A. Problem Formulation

We consider a surveillance network that expands through a sequence of T camera stages. At stage t, the learner receives only the data collected by the newly introduced camera or camera group,

$$\mathcal{D}_t = \{(x_i^t, y_i)\}_{i=1}^{N_t}, \qquad y_i \in \mathcal{Y}_t,$$

where x_i^t is a pedestrian image and $\mathcal{Y}_t$ is the set of identities observed at stage t. Unlike conventional class-incremental or lifelong ReID protocols with disjoint label spaces, CI-ReID allows repeated identities,

$$\mathcal{Y}_t \cap \mathcal{Y}_{<t} \neq \varnothing, \qquad \mathcal{Y}_{<t} = \bigcup_{s=1}^{t-1} \mathcal{Y}_s.$$

Thus, a newly deployed camera can contain both identities that have never been observed and identities that appeared in previous stages under different camera conditions.

Our primary benchmark follows a supervised CI-ReID protocol. Identity annotations belong to a globally consistent label space: if a person reappears in a later stage, the new samples retain the same identity label. This supervision provides cross-stage identity correspondence, but it does not permit historical replay. During stage t, the learner can access only the current data $\mathcal{D}_t$ and a compact state inherited from stage $t-1$, consisting of the model parameters θ_{t-1}, fixed geometric targets W_{t-1}, and stored identity prototypes $\mathcal{P}_{t-1}$. Raw images from $\{\mathcal{D}_s\}_{s<t}$ are unavailable.

The scope of this work is representation evolution after correspondence is given or estimated. Specifically, the objective is to learn a model f_{θ_t} that (1) integrates the new appearance evidence of repeated identities, (2) learns newly introduced identities, (3) preserves a stable global embedding geometry, and (4) generalizes to cameras that remain unseen during training. Acquiring correspondence without globally consistent annotations is a complementary upstream problem and is evaluated as an additional extension.

B. Overview

To address the challenge of identity representation evolving continuously with the expansion of cameras in CI-ReID, we propose the Camera-Incremental Geometric Representation Evolution framework,as shown in the Fig. 3. This framework centers on geometric representation learning, explicitly modeling the structural evolution of identity features in the embedding space to effectively absorb and integrate information from new perspectives. CIGRE consists of two key, mutually reinforcing modules: (1) **Dynamic Geometric Alignment** is used to maintain the stability of the global discriminative structure during camera increments.(2) **Cross-Camera Identity Consolidation** is used to gradually aggregate identity features observed from new cameras into existing identity representations, thereby alleviating the problem of identity fragmentation across stages. For each stage, CIGRE first identifies newly observed labels and expands the fixed target system. It then optimizes the current model using the baseline ReID objectives, cross-camera identity consolidation, and geometric alignment. After optimization, the baseline-specific model-consolidation rule is applied without modification. Finally, current-stage embeddings are extracted to initialize prototypes for new identities and update prototypes for repeated identities. This stage-wise design uses no historical raw images.

C. Dynamic Geometric Alignment

Traditional classification-based person re-identification methods typically rely on a linear classifier to learn the identity decision boundary [40], [46], [70]. The linear classifier can be represented as a weight matrix:

$$\mathcal{W} = [w_1, w_2, ...w_K] \in \mathcal{R}^{d\times K},$$

where d is the feature dimension, K is the number of identity categories, and w_k represents the category prototype corresponding to the k-th identity. In static or slowly changing scenarios, this paradigm can fit a stable class distribution by adjusting classification weights. However, in CI-ReID scenarios, with the continuous introduction of new camera perspectives, the feature distribution of the same identity will continue to expand and change, making it difficult for fixed or weakly updated linear classifiers to absorb newly exposed discriminative information, while over-updating can easily destroy the existing discriminative structure. To address this, we propose a DGA mechanism based on the the ETF classifier.

The ETF classifier constructs a set of geometrically well-defined class prototypes in the embedding space, ensuring that the angle between any two different class prototypes is equal, thereby forming a discriminative structure in the feature space that maximizes the class margin [71], [27]. Formally, the weight matrix of the ETF classifier at the current stage can be denoted as:

$$\mathcal{W} = [w_1, w_2, ...w_K],$$

Each column vector satisfies the unity norm constraint $\|w_k^{(t)}\|^2 = 1$, and any two fixed target vector satisfy the isogonality condition:

$$\langle w_i, w_j \rangle = -\frac{1}{K-1}, i \neq j.$$

This allows for the explicit characterization of the global geometric relationships between identities within the embedded space.

In the CI-ReID scenario, as new identities are continuously introduced, the number of categories K_t that the classifier needs to model increases with each stage. To address this, we allow the ETF classifier to dynamically expand with each stage, and incrementally model the newly added categories while maintaining the stability of the existing geometry.The specific expansion mechanism is shown in the Fig. 4. Formally, the weight matrix of the ETF classifier in stage t is:

$$\mathcal{W} = [w_1^{(t)}, w_2^{(t)}, ...w_{K_t}^{(t)}],$$

Each column vector satisfies the unity norm constraint $\|w_k^{(t)}\|^2 = 1$, and any two column vectors satisfy the isogonal condition:

$$\langle w_i^{(t)}, w_j^{(t)} \rangle = -\frac{1}{K_t - 1}, i \neq j,$$

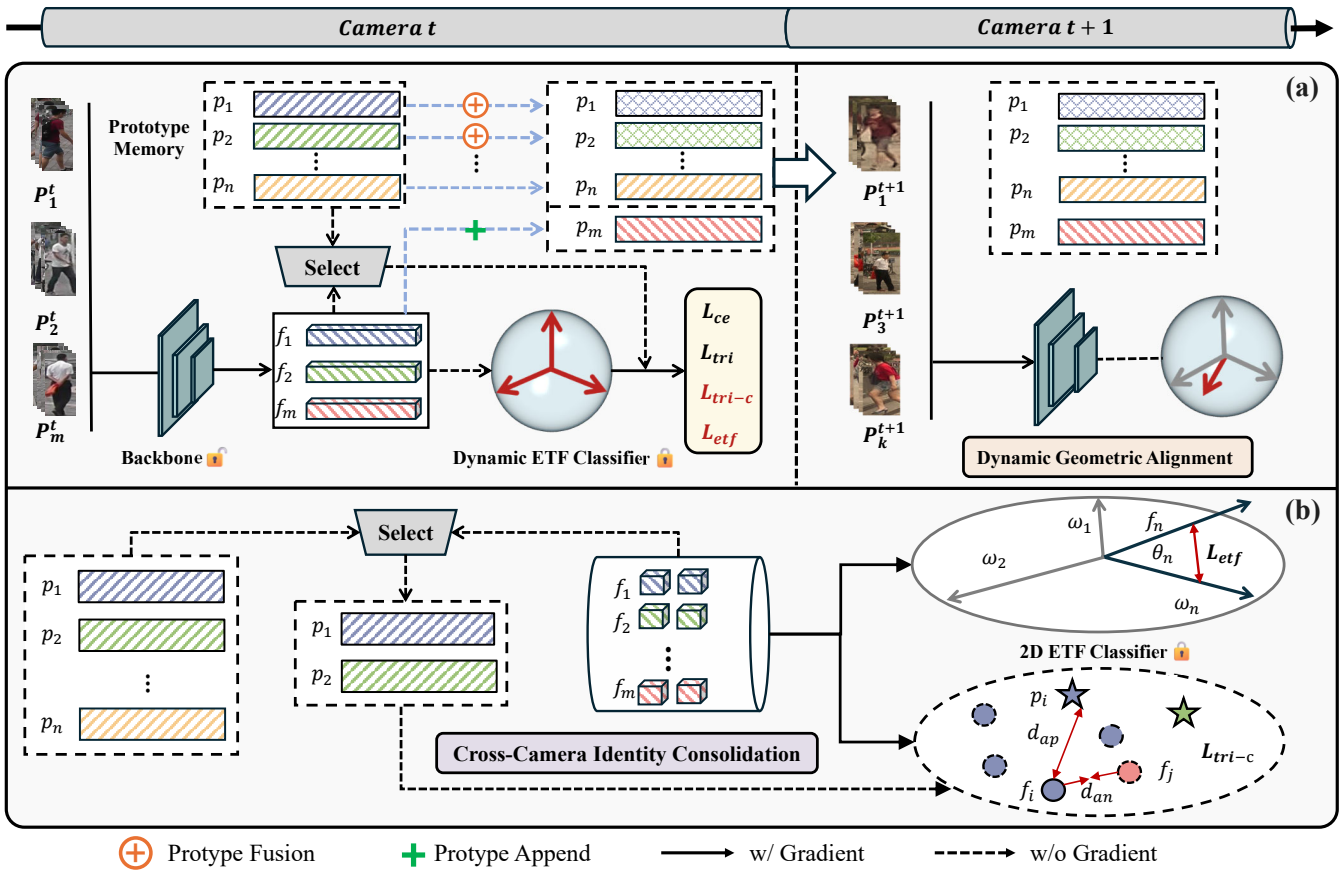


Fig. 3. An overview of the proposed CIGRE framework. (a) The overall process of dynamic identity evolution. Given an input image P_i^t from camera t, the backbone network extracts features f_i. The ETF classifier is frozen during training to provide a fixed, structured geometric target for feature learning. These features are then processed by the CCIC module, interacting with the global dynamic identity prototype $\{p_1, \ldots, p_n\}$. The classifier extends its geometry by appending a new target vector v_{m+1} to the existing isoangular compact framework. (b) An illustration of the CCIC strategy and ETF geometry diagram. Globally relevant prototypes are retrieved to ensure fragmented features are associated with these prototypes. The 2D ETF classifier demonstrates its underlying geometry: features are optimized to align with the corresponding fixed target vector by minimizing angular deviations, while L_{tri-c} simultaneously enforces global intra-class compactness (d_{ap}) and inter-class separability (d_{an}).

When a new identity category appears in stage $t+1$, we extend the classifier while keeping the existing ETF fixed target vector unchanged:

$$\mathcal{W}_{t+1} = [\mathcal{W}_{\sqcup}, w_{K_t+1}^{(t+1)}, w_{K_t+2}^{(t+1)}, \ldots w_{K_t+n}^{(t+1)}],$$

Where n is the number of new categories. The way the new vector is constructed depends on the relationship between the current number of categories and the embedding dimension.

When $K_t < d$,the embedding space still has a non-zero orthogonal complement space. We let the newly added vector lie in the orthogonal complement of the subspace spanned by the existing vectors:

$$w_{k_t+1}^{(t+1)} \in Null(W_t^{\top}), \qquad \|w_{k_t+1}^{(t+1)}\|^2 = 1,$$

This ensures that it is strictly orthogonal to all existing fixed target vectors and naturally maintains the geometric expansion property of ETFs. When $K_t > d$, Due to the limitation of the embedding space dimension, it is impossible to construct any new strictly orthogonal vectors. In this case, we construct unit vectors that are as orthogonal as possible to the existing vectors by solving the following optimization problem:

$$w_{k_t+1}^{(t+1)} = \arg min_{\|w\|^2=1} \sum_{k=1}^{K_t} \langle w, w_k(t) \rangle^2,$$

This preserves the geometric uniformity and discriminant structure of the original ETF classifier to the greatest extent possible.

Through the aforementioned dynamic expansion strategy, the ETF classifier can gradually incorporate new identities without disturbing existing fixed target vectors, providing a stable global reference for the geometric alignment of subsequent feature distributions.

D. Cross-Camera Identity Consolidation

While DGA provides a stable discriminative structure for the global feature space, identity-level representation still faces significant challenges in CI-ReID scenarios. Due to differences in viewpoint, imaging conditions, and background distribution among different cameras, features extracted for the same identity at different stages often exhibit large distribution

shifts. Without explicit identity-level constraints, the model is prone to forming fragmented identity representations during cross-stage learning, thereby weakening identity consistency.

To this end, we propose CCIC strategy. CCIC maintains a data-driven visual prototype $P_y^t \in \mathbb{R}^d$ for each identity observed up to stage t. The prototype is distinct from the fixed geometric target w_y: P_y^t summarizes visual evidence, whereas w_y specifies a stable class-level direction. After completing the optimization of stage t, we extract the embeddings of all currently available samples of identity y and compute its normalized current-stage center:

$$C_y^t = \text{Norm}\left(\frac{1}{N_y^t}\sum_{i:y_i=y}\text{Norm}\left(f_{\theta_t}(x_i^t)\right)\right),$$

where N_y^t is the number of current-stage samples of identity y. If y is observed for the first time, its prototype is initialized from visual evidence:

$$P_y^t = C_y^t, \qquad y \notin \mathcal{Y}_{<t}.$$

If y has appeared previously, its prototype is updated once at the stage boundary through normalized momentum consolidation:

$$P_y^t = \text{Norm}\left(mP_y^{t-1} + (1-m)C_y^t\right), \qquad y \in \mathcal{Y}_t \cap \mathcal{Y}_{<t},$$

Where $m \in [0,1)$ is the momentum coefficient, used to balance historical information with current observations.

During the loss calculation phase, for the identity set $\mathcal{Y}_b$ appearing in the current mini-batch, we select the corresponding prototype $\{P_y | y \in \mathcal{Y}_b\}$ from the global prototype library and incorporate it together with the current batch features into the calculation of the triplet loss.Specifically, given an anchor feature f_i, its positive sample set not only includes intra-batch features of the same identity, but also the corresponding global prototype P_{y_i}, thus yielding the identity integration triplet loss in the following form:

$$\mathcal{L}_{tri-c} = \sum_i [max(\mathcal{D}(f_i, P_{y_i}) - \mathcal{D}(f_i, f_j^-) + \alpha, 0],$$

Where f_j^- represents negative sample features from different identities, $\mathcal{D}(\cdot,\cdot)$ is the cosine distance, and α is the margin hyperparameter. By introducing a global prototype as a stable positive sample reference, this loss effectively constrains identity consistency across stages and cameras.

Furthermore, to further enhance the consistency between the identity prototype and the global geometric structure, we introduce a joint ETF alignment loss $\mathcal{L}_{etf}$ for prototype and instance. Specifically, for the features f_y and corresponding the global prototype P_y of identity y, we constrain it to maintain consistency with the corresponding unit vector w_y in the ETF classifier in the angular space:

$$\mathcal{L}_{etf} = \sum_{y\in\mathcal{Y}_b}[(1 - cos(f_y, w_y)) + (1 - cos(P_y, w_y))],$$

Where $cos(\cdot,\cdot)$ represents the cosine similarity. w_y represents the unit targets vector corresponding to identity y in the ETF classifier. This constraint ensures that the identity-level prototypes not only maintain consistency in the time dimension, but also align with the global discriminative geometry defined by the ETF.

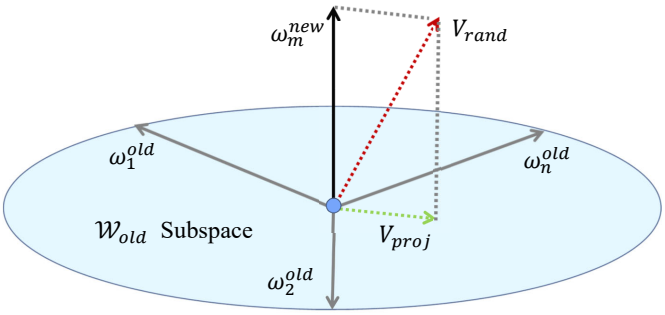


Fig. 4. An illustration of the orthogonalization implementation in the dynamic ETF classifier expansion. We maintain n fixed target vectors W_n^{old} in the m-dimensional subspace $\mathcal{W}_{old}$. When a new class is detected, a candidate vector V_{rand} is randomly initialized. Its orthogonal projection onto W_{old} is calculated. After normalization and scaling, the selected component becomes the new fixed target vector W_{new}. This mechanism effectively expands the classifier's geometry and ensures the orthogonality between W_{new} and W_{old}.

Through the above design, CCIC can stably incorporate the identity features exposed by new cameras into the existing identity representation without the constraint of historical data, effectively alleviating the problem of identity fragmentation in cross-stage learning, and working with DGA to promote the geometric evolution of identity representation.

E. Overall Training Objectives

Based on the modeling and loss definition, we constructed a unified training objective to simultaneously achieve identity discrimination. Specifically, the overall optimization process is driven by four complementary loss terms. To establish basic identity recognition capabilities, we adopt cross-entropy loss ($\mathcal{L}_{ce}$) [17] and triplet loss ($\mathcal{L}_{tri}$), which are standard losses in the ReID task. Simultaneously, by combining the CCIC objective with the ETF alignment loss, the consistency of features in geometric space is explicitly constrained at both the instance and identity prototype levels. This enables the model to effectively absorb new camera information and achieve continuous aggregation and alignment of cross-camera identity representations.

The overall training objective is formulated as:

$$\mathcal{L} = \mathcal{L}_{ce} + \lambda_1\mathcal{L}_{tri} + \lambda_2\mathcal{L}_{tri-c} + \lambda_3\mathcal{L}_{etf},$$

where λ_1 follows the settings from the original baseline method, λ_2 and λ_3 balance the contributions of identity consolidation and geometric alignment, respectively. By jointly optimizing these objectives, the model can continuously improve identity representation, effectively incorporate emerging camera-specific information, and maintain a consistent global geometry as the camera system expands over a long period. For the training process of our framework, please refer to Algorithm 1 for detailed steps.

The model-consolidation operation in Algorithm 1 is inherited from the selected lifelong ReID baseline. For an EMA-based baseline, a generic consolidation rule can be written as

$$\theta_t \leftarrow \rho\theta_{t-1} + (1-\rho)\theta_t^{\text{new}},$$

TABLE I
CAMERA INTRODUCTION ORDERS ON MARKET-1501 AND MSMT17.MARKET-1501'S C_3 AND MSMT17'S C_1, C_2, AND C_9 ARE UNSEEN CAMERAS STRATEGICALLY SELECTED BASED ON THEIR SAMPLE SPARSITY.

Order	Camera Sequence on Market-1501	Camera Sequence on MSMT17
Order 1	$C_0 \to C_1 \to C_2 \to C_4 \to C_5$	$C_0 \to C_3 \to C_4 \to C_5 \to C_6 \to C_7 \to C_8 \to C_{10} \to C_{11} \to C_{12} \to C_{13} \to C_{14}$
Order 2	$C_2 \to C_0 \to C_1 \to C_5 \to C_4$	$C_6 \to C_{10} \to C_{11} \to C_5 \to C_3 \to C_{13} \to C_7 \to C_0 \to C_8 \to C_{14} \to C_{12} \to C_4$
Order 3	$C_4 \to C_2 \to C_0 \to C_5 \to C_1$	$C_{14} \to C_{13} \to C_{12} \to C_{11} \to C_{10} \to C_8 \to C_7 \to C_6 \to C_5 \to C_4 \to C_3 \to C_0$

Algorithm 1 Training Pipeline of CIGRE

Input: Camera-stage data $\{\mathcal{D}_t\}_{t=1}^{T}$; initial model θ_0; momentum m
Output: Final ReID model θ_T

Initialize target memory $W_0 \leftarrow \emptyset$ and prototype memory $\mathcal{P}_0 \leftarrow \emptyset$
for $t = 1$ to T **do**
  Obtain current identity set $\mathcal{Y}_t$ from $\mathcal{D}_t$
  $\Delta\mathcal{Y}_t \leftarrow \mathcal{Y}_t \setminus \mathcal{Y}_{<t}$
  Expand W_{t-1} only for identities in $\Delta\mathcal{Y}_t$ using DGA; freeze all targets
  Initialize the current learner from the baseline model at stage $t-1$
  for each identity-balanced mini-batch (x, y) from $\mathcal{D}_t$ **do**
    Extract normalized embeddings $f = f_{\theta_t}(x)$
    Retrieve P_y^{t-1} for repeated identities when available
    Compute $\mathcal{L}_{\text{ce}}$, $\mathcal{L}_{\text{tri}}$, $\mathcal{L}_{\text{tri-c}}$, and $\mathcal{L}_{\text{etf}}$
    Update the current learner by back-propagation
  end for
  Apply the baseline-specific model-consolidation rule to obtain θ_t
  Extract current-stage embeddings using f_{θ_t} and compute C_y^t for each $y \in \mathcal{Y}_t$
  for each $y \in \mathcal{Y}_t$ **do**
    if P_y^{t-1} does not exist **then**
      $P_y^t \leftarrow C_y^t$
    else
      $P_y^t \leftarrow \text{Norm}(mP_y^{t-1} + (1-m)C_y^t)$
    end if
  end for
end for
return θ_T

CIGRE does not alter the baseline-specific coefficient or strategy. Its consistent improvements when integrated with different baselines demonstrate compatibility with different consolidation mechanisms.

IV. EXPERIMENTS

A. Experimental Setup

Datasets and Protocols. We evaluate our method on two prominent ReID benchmarks: Market-1501 [72] and MSMT17 [73]. To simulate camera-network expansion, each dataset is partitioned by camera. Market-1501 camera C3 is reserved as an unseen camera, while MSMT17 cameras C1, C2, and C9 are reserved as unseen cameras because of their relatively sparse samples. The remaining cameras are introduced sequentially according to the three orders listed in Table I. Each order is trained independently from scratch.

The primary evaluation follows supervised CI-ReID. Identity annotations are globally consistent across cameras and stages, so the label of a repeated identity directly supplies its cross-stage correspondence. At stage t, training uses only images from the newly introduced camera, together with model parameters, fixed geometric targets, and identity prototypes retained from previous stages. Historical raw images are neither accessed nor stored. Unless otherwise stated, all main results are reported as the mean and standard deviation over the same three camera orders.

The identity-overlap ratio at stage t is defined as

$$r_t = \frac{|\mathcal{Y}_t \cap \mathcal{Y}_{<t}|}{|\mathcal{Y}_t|}.$$

Under the adopted Market-1501 camera sequence, the natural overlap ratio is 0.847

Evaluation Metrics. We report mean average precision (mAP) and Rank-1 accuracy (R@1). Seen-camera evaluation measures retrieval performance on cameras introduced up to the current stage, whereas unseen-camera evaluation measures zero-shot generalization to the held-out cameras. Stage-wise curves are used as diagnostics for one explicitly specified camera order. Tables II and III aggregate independent runs from all three orders and report mean $\pm$ standard deviation.

Implementation Details. We used ResNet-50 [74] as the backbone network and optimize it with SGD. The first camera stage is trained for 80 epochs, and each subsequent stage is trained for 60 epochs. We use a RandomIdentitySampler with $P = 32$ identities and $K = 4$ images per identity, giving a batch size of 128. The learning rate is 3.5×10^{-4} with the same decay schedule for all compared methods. The model-consolidation rule and its associated update coefficients follow the original baseline method without modification; CIGRE does not introduce an additional model-fusion strategy. For CCIC, the momentum coefficient used to consolidate a stored prototype with the current-stage identity center is set to $m = 0.5$. The baseline triplet coefficient is $\lambda_1 = 1.5$, and the CIGRE coefficients are $\lambda_2 = 1.5$ and $\lambda_3 = 1.5$. The same λ_2 and λ_3 are used on Market-1501 and MSMT17; MSMT17 is not separately retuned for the main comparison.

B. Comparison with State-of-the-Art

To validate the effectiveness of the proposed framework, we integrate CIGRE into two baseline methods, PKA [1] and DKC [75], and compare it with several mainstream LReID methods, including DKC, PKA, DKP [76], LSTKC [77], and

TABLE II
QUANTITATIVE COMPARISON WITH STATE-OF-THE-ART METHODS ON MARKET-1501 AND MSMT17. THE BEST PERFORMANCE IS **BOLDED**, AND THE SUB-OPTIMAL PERFORMANCE IS UNDERLINED. ± FOLLOWED BY THE VARIANCE OF THREE DIFFERENT CAMERA ORDER RUNS.

	Method	Publication	Seen-Avg on Market-1501		Unseen-Avg on Market-1501		Seen-Avg on MSMT17		Unseen-Avg on MSMT17	
			mAP	R@1	mAP	R@1	mAP	R@1	mAP	R@1
CIL	LwF	TPAMI 2017	19.4 ± 1.0	25.8 ± 1.0	26.4 ± 0.6	40.3 ± 0.6	10.9 ± 1.0	20.8 ± 1.2	5.2 ± 1.1	11.3 ± 2.3
	SPD	ICCV 2019	18.7 ± 0.9	25.0 ± 1.1	26.2 ± 0.5	39.8 ± 0.5	5.0 ± 0.9	10.6 ± 1.1	4.3 ± 0.5	8.9 ± 1.4
	PRD	ICML 2023	12.3 ± 0.5	15.4 ± 1.3	15.4 ± 0.6	20.2 ± 0.7	4.8 ± 1.2	8.4 ± 1.3	3.4 ± 0.6	5.6 ± 0.7
	PRAKA	ICCV 2023	25.1 ± 1.2	45.8 ± 1.2	32.6 ± 1.1	45.2 ± 1.2	13.0 ± 1.7	27.2 ± 1.9	8.6 ± 1.4	15.3 ± 1.8
LReID	DKP	CVPR 2024	37.6 ± 0.3	60.3 ± 0.4	51.9 ± 0.9	78.9 ± 0.2	19.3 ± 2.0	42.2 ± 2.8	15.1 ± 1.5	25.1 ± 2.4
	LSTKC	AAAI 2024	39.8 ± 0.7	62.7 ± 0.8	53.6 ± 2.0	80.5 ± 0.9	18.9 ± 1.0	44.3 ± 1.7	15.1 ± 1.1	26.7 ± 0.7
	LSTKC++	TPAMI 2025	39.7 ± 1.0	61.4 ± 0.8	56.1 ± 0.9	81.3 ± 1.8	18.5 ± 0.6	40.8 ± 0.6	13.5 ± 0.2	24.6 ± 1.8
	PKA	IJCAI 2025	37.7 ± 1.5	61.0 ± 1.5	51.9 ± 1.7	79.4 ± 1.3	18.5 ± 0.6	41.7 ± 0.2	13.7 ± 0.5	24.5 ± 1.2
	PKA+CIGRE	Ours	44.2 ± 1.0	67.3 ± 0.9	57.7 ± 0.9	82.5 ± 1.8	**20.3 ± 0.8**	**45.3 ± 1.1**	**16.0 ± 1.2**	**28.8 ± 2.9**
	DKC	CVPR 2025	37.8 ± 0.2	60.1 ± 0.2	51.8 ± 1.4	78.7 ± 1.0	19.4 ± 1.8	42.5 ± 2.5	14.6 ± 1.6	24.1 ± 2.7
	DKC+CIGRE	Ours	**45.5 ± 0.7**	**67.6 ± 0.8**	**58.0 ± 0.9**	**83.1 ± 0.3**	20.1 ± 1.0	44.4 ± 1.3	15.0 ± 1.4	26.8 ± 2.2

LSTKC++ [78]. Furthermore, we introduce several typical CIL methods for a comprehensive comparison, such as LwF [79], SPD [80], PRAKA [12], and PRD [81]. Specifically, we statistically analyzed the average performance of each method under different camera introduction orders. We reported the average results and standard deviations for all seen and unseen domains, thereby comprehensively evaluating the stability and generalization of model as the number of cameras increases. To ensure a fair comparison, all methods employ the same backbone network architecture and a unified training strategy. For the CIL method, we use the widely used triplet loss [82] for training to ensure consistency and comparability with the LReID method.Specific experimental results are shown in the Table II.

Performance of CIL Methods. We systematically compared the proposed framework with several representative CIL methods. Experimental results show that the performance of traditional CIL methods in person ReID tasks is significantly limited. This performance gap stems from the fundamental difference in their learning paradigms. Traditional CIL methods are usually based on the assumption of a closed class hierarchy, and its core lies in learning and maintaining stable discriminative boundaries of known classes. However, ReID is essentially a metric learning task in an open-set scenario, aiming to construct a discriminative embedding space represented by the relative distance between features, rather than relying on a fixed classification decision surface. Therefore, the strict classification boundary emphasized by CIL has clear limitations in the ReID scenario, which weakens the model's ability to generalize and retrieve identity features.

Seen-Camera Performance Evaluation. We compare the proposed method with representative LReID approaches on the seen-camera setting over Market-1501 and MSMT17, with results summarized in Table II. Before incorporating CIGRE, our PKA baseline exhibit suboptimal performance under seen-camera evaluation. This limitation primarily arises from the inability of this method to effectively characterize the cross-camera feature evolution of the same identity. In particular, it lack an explicit mechanism to reconcile view-induced discrepancies, leading to fragmented identity representations andand thereby reducing discriminability. With the integration of CIGRE, both PKA and DKC consistently achieve substantial improvements across all evaluation metrics, surpassing current SOTA LReID methods. Taking PKA as an example, on Market-1501 and MSMT17, the mAP/R@1 under the seen-avg protocol is improved by 6.5%/6.3% and 1.8%/3.6%, respectively. Furthermore, compared with the strongest existing LReID competitor, our method yields additional gains of 4.4%/4.6% and 0.9%/2.8%. These results indicate that CIGRE effectively mitigates cross-camera representation inconsistency.

To assess knowledge accumulation and forgetting during camera-incremental training, we evaluate model performance after each stage. As shown in Fig. 5(a), which reports Market-1501 Order 1, CIGRE is only marginally better than PKA at the first stage and slightly trails some competing methods such as LSTKC++. This is expected because, with observations from only a limited number of cameras, the global geometric structure has not yet been sufficiently established.

As more cameras are introduced, CIGRE quickly gains a clear advantage. From stage 2 onward, PKA+CIGRE surpasses the compared methods and maintains its lead throughout the remaining stages. The additional camera views provide richer cross-camera evidence for prototype consolidation and geometric alignment, enabling CIGRE to better preserve intra-class consistency and inter-class separation as the representation space evolves. These results indicate that CIGRE requires a short initialization period, after which its geometric representation becomes increasingly effective in mitigating representation drift and preserving previously acquired knowledge.

Regarding the performance drop in the fifth stage, the camera difficulty heatmap shown in Fig. 6 confirms the challenging nature of this specific camera. We utilized a pre-trained, frozen ResNet-50 to extract features and measured

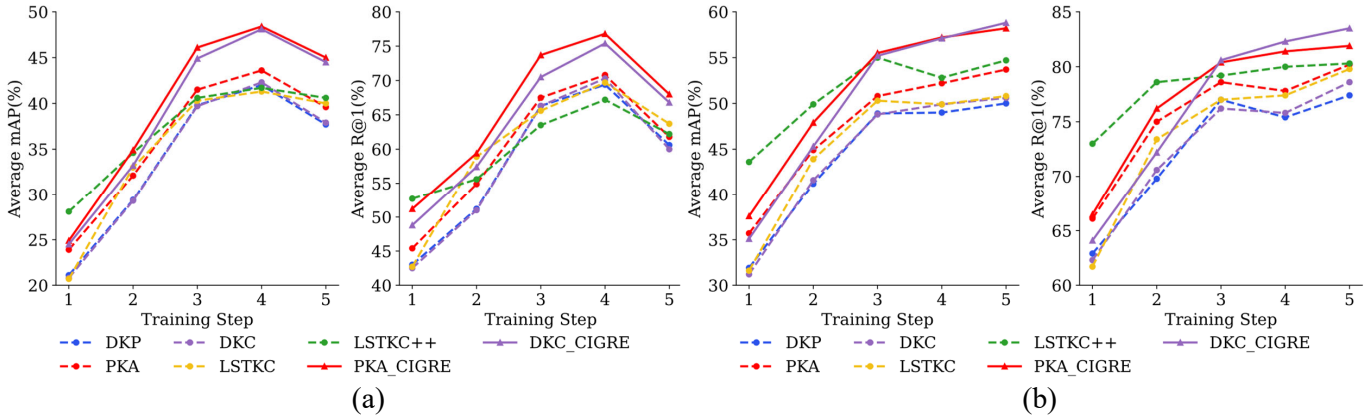


Fig. 5. Performance trends on the Market-1501: (a) Seen cameras performance after each training stage. (b) Unseen cameras performance after each training stage.

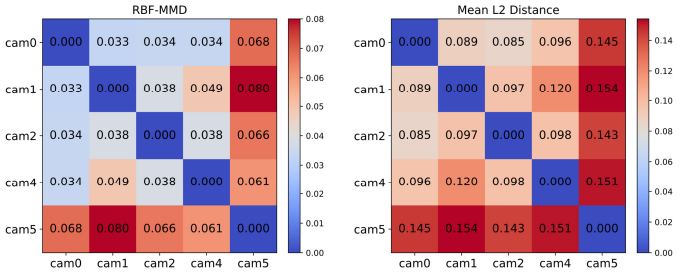


Fig. 6. Camera difficulty heatmap measured on the cross-camera test set.

inter-camera discrepancies using RBF-MMD and average L2 distance. Compared to C0–C4, C5 exhibits RBF-MMD values ranging from 0.061 to 0.080 (with a mean of 0.0688), representing the greatest overall discrepancy in the sequence. Although the introduction of C5 led to a performance decline across nearly all methods, CIGRE maintained its lead; notably, its performance on the unseen camera remained unaffected during this stage, demonstrating superior robustness without compromising cross-camera generalization capabilities.

Unseen-Camera Generalization Evaluation. To evaluate generalization to unseen camera environments, we compare different methods under the unseen-avg protocol. Existing methods generally suffer noticeable performance degradation on long unseen-camera sequences, particularly on the more challenging MSMT17 dataset. Representative LReID methods such as DKP and LSTKC++ achieve only about 13%–15% mAP on unseen cameras, suggesting that without explicit geometric or structural constraints, their representations are more vulnerable to camera-distribution shifts. In contrast, CIGRE consistently achieves stronger unseen-camera performance across the evaluated metrics. This improvement is mainly attributed to its stable global representation structure: DGA and the associated geometric constraints encourage the model to preserve a structured feature space that is less dependent on specific camera domains. As a result, even for unseen identities and camera views, CIGRE better maintains feature-space consistency and alleviates the representation distortion introduced by continual camera expansion.

To further examine how unseen-camera generalization evolves during camera-incremental training, we report the unseen-avg performance after each stage in Fig. 5(b). At the first stage, although the model is trained with only a limited number of camera views, the CIGRE-based model already shows favorable cross-camera transferability, indicating that its initial representation space possesses useful generalization potential. As more cameras are introduced, this advantage becomes increasingly evident. Existing LReID methods tend to encounter a generalization bottleneck after the third incremental stage and exhibit larger performance fluctuations, suggesting increasing sensitivity to the accumulated seen-camera distributions. In contrast, CIGRE shows a more stable and approximately linear improvement trend throughout the incremental process. This behavior reflects the structural consistency maintained by CIGRE, which helps preserve transferable identity representations as the camera network evolves. Overall, the results demonstrate that CIGRE not only improves generalization to unseen cameras but also maintains this advantage throughout continual camera expansion, which is particularly desirable for practical deployment in dynamically growing camera networks.

C. Experiments Without Cross-Camera Global Identity Labels

Our primary protocol uses globally consistent identity annotations to isolate the problem of cross-camera representation evolution. To evaluate whether CIGRE critically depends on such oracle cross-camera links, we additionally consider a camera-local supervised protocol in which identity annotations are available only within each camera. Each incoming camera has its own local identity label space, and no global identity mapping is provided across cameras or incremental stages.

The local identities in the first camera initialize the global identity space and historical prototype memory. For a subsequent camera at stage t, we freeze the model inherited from stage $t-1$ and use it to extract features for the current images. Let $\ell \in \mathcal{L}_t$ denote a camera-local identity and $\mathcal{I}_{t,\ell}$ its image set. Its current local prototype is computed as

TABLE III
AN ABLATION STUDY OF THE CORE COMPONENTS OF OUR FRAMEWORK ON MARKET-1501 AND MSMT17 USING PKA AS A BASELINE.

Dynamic ETF	L_{etf}	CCIC	Seen-Avg on Market-1501		Unseen-Avg on Market-1501		Seen-Avg on MSMT17		Unseen-Avg on MSMT17	
			mAP	R@1	mAP	R@1	mAP	R@1	mAP	R@1
			37.7±1.5	61.0±1.5	51.9±1.7	79.4±1.3	18.5±0.6	41.7±0.2	13.7±0.5	24.5±1.2
✓			39.3±1.7	62.5±1.6	52.8±2.3	79.0±1.6	16.5±1.1	39.8±1.5	12.3±1.2	22.7±3.0
✓	✓		40.8±1.0	64.1±0.4	54.7±1.1	81.2±0.5	17.1±1.1	40.8±1.5	13.4±1.6	24.7±3.6
		✓	43.3±1.2	66.2±1.0	56.4±0.9	82.1±1.1	19.4±0.8	44.7±1.0	15.1±1.4	27.6±2.7
✓	✓	✓	**44.2±1.0**	**67.3±0.9**	**57.7±0.9**	**82.5±1.8**	**20.3±0.8**	**45.3±1.1**	**16.0±1.2**	**28.8±2.9**

$$C_{t,\ell} = \mathrm{Norm}\left(\frac{1}{|\mathcal{I}_{t,\ell}|}\sum_{x\in\mathcal{I}_{t,\ell}} \mathrm{Norm}\left(f_{\theta_{t-1}}(x)\right)\right).$$

We then compare $C_{t,\ell}$ with every stored historical prototype P_g^{t-1}, where $g \in \mathcal{G}_{t-1}$ is an existing global identity, using cosine similarity.

$$s_{\ell g} = \cos\left(C_{t,\ell}, P_g^{t-1}\right).$$

For each current local identity, we find its nearest historical identity,

$$g^\star(\ell) = \arg\max_{g\in\mathcal{G}_{t-1}} s_{\ell g},$$

and, in the reverse direction, find the nearest current local identity for each historical identity,

$$\ell^\star(g) = \arg\max_{\ell\in\mathcal{L}_t} s_{\ell g}.$$

The current identity ℓ is assigned the existing global label $g^\star(\ell)$ only if the pair is mutually nearest,

$$\ell^\star(g^\star(\ell)) = \ell.$$

Otherwise, ℓ is treated as a newly observed identity and assigned a new global label. This mutual-nearest-neighbor rule produces conservative one-to-one cross-stage matches without using cross-camera ground-truth annotations.

Table IV shows that CIGRE remains effective when cross-stage correspondence is estimated rather than provided by global identity annotations. Across both PKA and DKP, CIGRE consistently improves seen-camera performance and also benefits most unseen-camera metrics. The only exception is the unseen-camera R@1 of DKP+CIGRE, which decreases despite the improvement in unseen-camera mAP. This suggests that noisy or uncertain cross-stage associations may affect ranking precision and retrieval coverage differently.

Overall, these results indicate that CIGRE is not restricted to oracle cross-stage identity labels. Given automatically estimated correspondences, DGA can still maintain an expandable global target structure, while CCIC can consolidate matched camera-specific identity fragments. At the same time, the remaining gap from the primary supervised setting and the metric variation observed with DKP confirm that correspondence quality remains an important upstream factor. Therefore,

TABLE IV
PERFORMANCE ON MARKET-1501 USING CAMERA-LOCAL IDENTITY ANNOTATIONS AND MUTUAL-NEAREST-NEIGHBOR CROSS-STAGE CORRESPONDENCE.

Method	Seen-Avg		Unseen-Avg	
	mAP	R@1	mAP	R@1
PKA	30.1	52.1	44.7	73.8
PKA+CIGRE	**39.1**	**63.0**	**53.1**	**78.6**
DKP	31.7	54.0	44.3	77.4
DKP+CIGRE	37.9	60.5	46.7	72.6

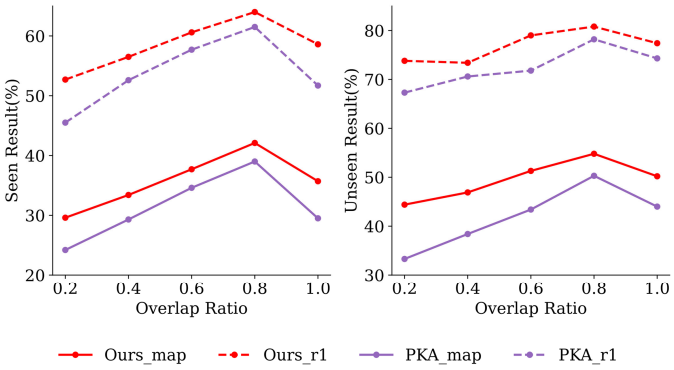


Fig. 7. Sensitivity to the identity overlap on seen and unseen cameras.

this experiment demonstrates the compatibility of CIGRE with estimated cross-stage correspondence while also highlighting the potential benefit of more accurate association strategies.

D. Sensitivity to Identity Overlap

Identity overlap distinguishes CI-ReID from conventional lifelong ReID protocols with disjoint identity sets. We construct controlled variants with different overlap ratios, where the ratio at a newly introduced camera is the proportion of its identities that have appeared in previous stages. The camera order, stage structure, and optimization settings are fixed, and PKA and CIGRE use exactly the same split at each ratio.

Changing the overlap ratio also changes the numbers of repeated identities, newly introduced identities, and corresponding training images. Therefore, the absolute curve should not be interpreted as a strictly controlled or monotonic measure of task difficulty. At a lower ratio, fewer repeated identities are available for cross-camera prototype consolidation; at a higher ratio, more identities must be updated under the new domain and can introduce stronger cross-stage interference. The more reliable comparison is the gap between CIGRE and PKA

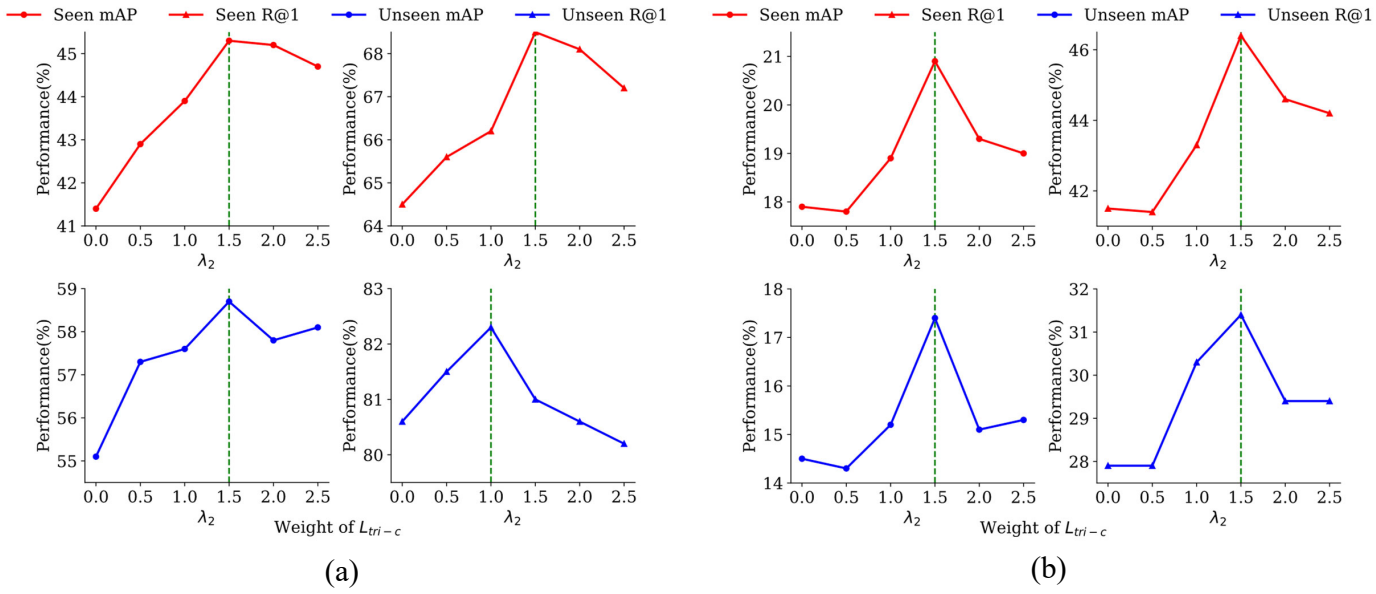


Fig. 8. Sensitivity of CIGRE to λ_2: (a) Results under the Order 1 setting on Market-1501. (b) Results under the Order 1 setting on MSMT17. The dashed lines indicate the optimal configuration for each evaluation metric.

under the same split. As shown in Fig. 7, CIGRE consistently outperforms PKA across all tested overlap regimes, although the magnitude of the improvement is not monotonic. These results show that the benefit of CIGRE is not restricted to one particular overlap setting.

E. Ablation Studies

To evaluate the effectiveness of each proposed component, we conduct ablation studies on both Market-1501 and MSMT17 datasets, taking PKA as the baseline.

Influence of Different Model Components. To verify the effectiveness of the core components in the proposed framework, we decomposed CIGRE into three parts. Dynamic ETF, its accompanying loss function L_{etf}, and the CCIC module. As shown in Table III, the baseline model struggles to maintain the stability of its representation as cameras are continuously introduced due to the lack of structural constraints in the feature space. First, we evaluated the effect of the dynamic ETF module on feature space reshaping. Introducing dynamic ETF significantly improved the model's performance on the Market-1501 dataset. Geometrically, dynamic ETF forces features of different identities to be uniformly distributed on a hypersphere by using predefined target vectors as class centers, maximizing inter-class separability. However, it is worth noting that on the more complex MSMT17 dataset, simply using dynamic ETF resulted in a performance decline. This indicates that, in the absence of explicit evolutionary constraints, forcibly mapping high-dimensional features to a stringent geometric target can cause structural perturbations in the feature distribution, making it difficult for the model to directly adapt to overly rigid geometric priors. To address the aforementioned limitations, we introduce the L_{etf} loss function to guide the smooth evolution of the manifold. The addition of L_{etf} improves the consistency across all metrics, further enhancing seen-avg mAP by 2.0% and 1.0% on Market-1501 and MSMT17, respectively. This improvement validates the necessity of manifold evolution continuity. L_{etf}, as a soft constraint mechanism, bridges the gap between features and the rigid ETFs objective, guiding the feature space to smoothly align during the incremental phase, thereby effectively mitigating the optimization difficulties and feature instability issues caused by simply using ETFs. Furthermore, we validated the role of the CCIC module in handling feature fragmentation. The introduction of this module significantly improved model performance, demonstrating that explicitly aggregating the scattering features of the same pedestrian under different cameras can effectively suppress viewpoint noise and enhance the robustness of the representation, making the model more discriminative in the face of cross-camera distribution shifts. Finally, when all components work together, the framework achieves optimal performance across all metrics. This cumulative performance improvement reveals a strong synergy among the three components. Dynamic ETF lays a solid geometric foundation, L_{etf} ensures smooth and stable representation evolution, and CCIC is responsible for integrating fragmented identity information. Together, they construct a robust framework that effectively addresses the challenges of representation drift and recognition in CI-ReID.

Influence of Hyperparameters. Fig. 8, Fig. 9 report the sensitivity of PKA+CIGRE to λ_2 and λ_3 on Market-1501 and MSMT17. The same preset values, $\lambda_2 = 1.5$ and $\lambda_3 = 1.5$, are adopted in all main experiments on both datasets. For the sensitivity study, we vary one coefficient at a time while fixing the other at 1.5 and keeping the camera order and optimization settings unchanged, so that the effect of each term can be examined independently.

For λ_2, both datasets exhibit a similar unimodal trend. When λ_2 is too small, the cross-camera consolidation constraint is insufficient, and fragmented representations of the same identity cannot be effectively aligned across incremental

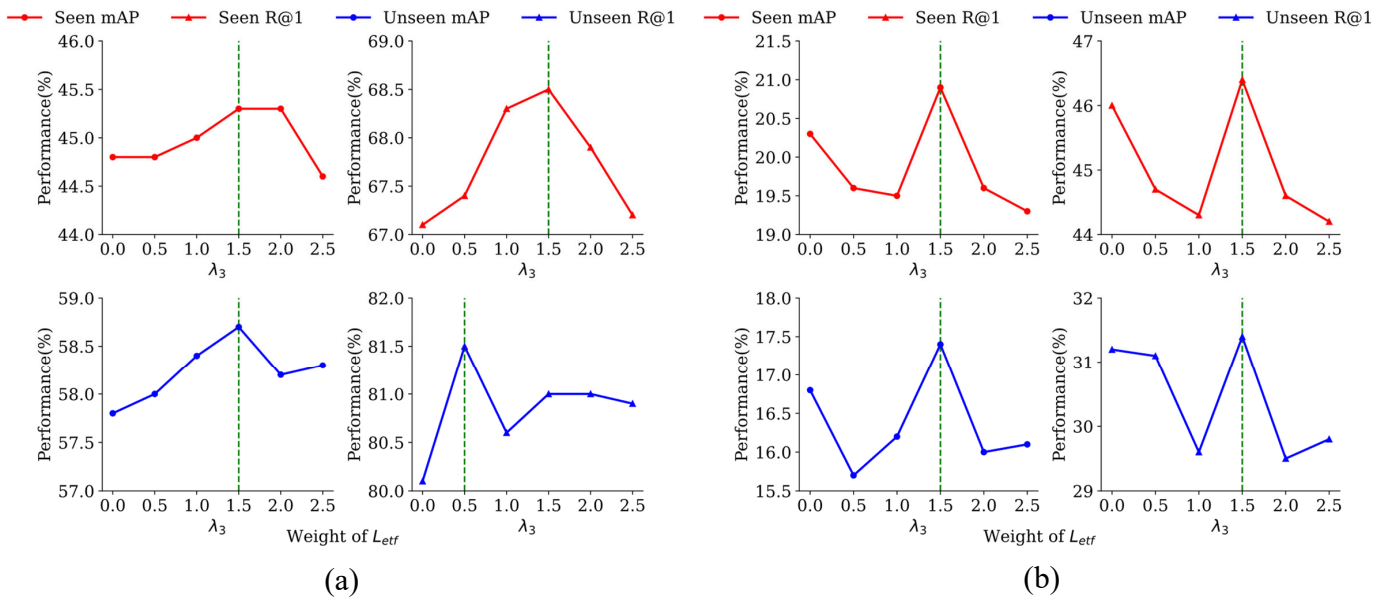


Fig. 9. Sensitivity of CIGRE to λ_3: (a) Results under the Order 1 setting on Market-1501. (b) Results under the Order 1 setting on MSMT17. The dashed lines indicate the optimal configuration for each evaluation metric.

cameras. Increasing λ_2 strengthens identity consistency and leads to steady performance improvements. However, when λ_2 becomes overly large, the model places excessive emphasis on historical consistency, which suppresses adaptation to newly observed appearances and eventually degrades retrieval performance. This effect is more pronounced on MSMT17, where the larger identity space and stronger cross-camera distribution variation make the stability–plasticity trade-off more sensitive. In particular, increasing λ_2 from 0 to 1.5 improves seen-camera mAP from approximately 17.9% to 20.9%, while further increasing it causes a noticeable decline.

In comparison, the model is less sensitive to λ_3 on both datasets. This coefficient controls the strength of the geometric structure constraint. Within a moderate range, the ETF-inspired prior regularizes the embedding space and helps preserve inter-class separation as the identity space expands. Since the geometric target structure itself already provides a relatively stable reference, moderate changes in λ_3 have a limited effect on the final performance. Only when λ_3 becomes excessively large does the structural prior begin to dominate the data-driven objective, reducing the flexibility of the representation in fitting complex appearance variations. This behavior is also observed on MSMT17, where the four evaluation metrics vary within approximately 2.2 percentage points over the tested range.

Overall, λ_2 mainly controls the balance between cross-camera consolidation and adaptation, whereas λ_3 regulates the strength of global geometric structure preservation. Importantly, the same preset value of 1.5 provides the best overall operating point for both coefficients on Market-1501 and MSMT17. This consistency suggests that although MSMT17 is more sensitive to the consolidation strength, the proposed method does not require dataset-specific retuning of the default hyperparameters used in the main experiments.

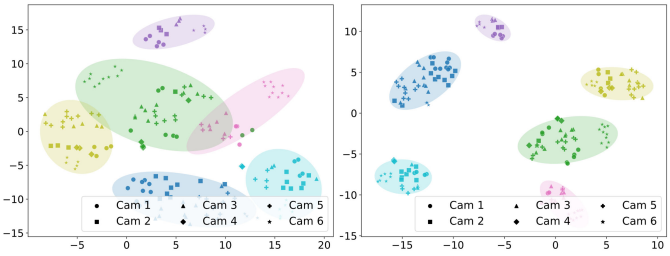


Fig. 10. Visual analysis of the feature embedding space: colors represent different IDs, and symbols represent different camera views. Compared to the baseline method (left), our method (right) achieves stronger intra-class aggregation and clearer inter-class separation in cross-camera scenarios.

F. Analysis

Embedding Space Analysis. The distribution structure of the feature space is a key metric for evaluating ReID models [83], [84], [85], [86]. To this end, we employ t-SNE to map high-dimensional features to a two-dimensional plane and compare the representation differences between baseline methods and our proposed method in cross-camera scenarios. Specifically, we select feature samples from multiple identities and distinguish and label the feature prototypes from different cameras to more clearly characterize the cross-view distribution. The final embedding space distribution is shown in the Fig. 10. Experimental observations show that the baseline model exhibits a relatively discrete feature distribution. Its learned embedding space is affected by camera view bias, showing dispersion across camera subclusters. Thanks to the strong local discriminative power of the linear classifier, the baseline model still demonstrates good performance. However, this representation, which relies on specific boundaries, is highly susceptible to feature drift and overlapping of old and new class discriminations during incremental updates. In contrast, the CIGRE framework aims to construct a stable

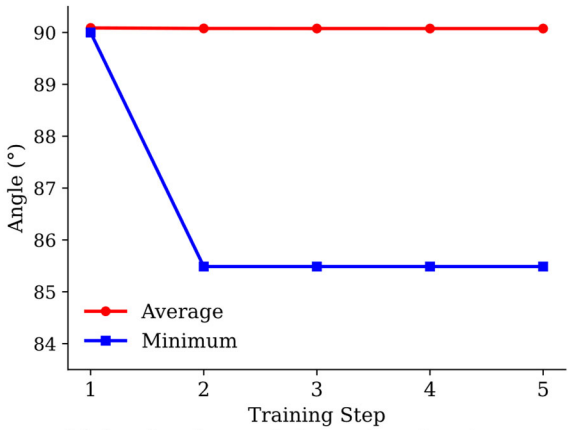


Fig. 11. The evolution of the angular relationship between target vectors during the training phase.

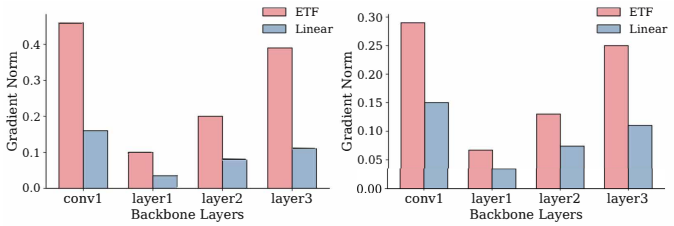


Fig. 12. Comparison of gradient norms across network layers at epoch 30 (left) and epoch 50 (right). Larger gradient norms indicate that the proposed framework delivers stronger supervisory signals during optimization, effectively mitigating gradient vanishing and thereby improving feature learning capacity and cross-scene generalization.

geometric discriminative space. By fusing the DGA and CCIC modules, the model not only significantly reduces identity feature shifts across camera views and achieves complete aggregation within classes, but also ensures inter-class distribution distances through a stable geometric space. This not only enhances the model's robustness in continuously incremental scenarios but also improves its generalization ability in unknown camera views.

Analysis of Gradient Dynamics. To further investigate the impact of our proposed frame on the model training process, we monitored and compared the gradient norms generated by the baseline method and our proposed method at each network layer. As shown in the Fig. 12, the experimental results show that our proposed method maintains a significantly higher gradient norm across all layers during the training phase. In conventional linear classification layers, the optimization process tends to favor learning relatively easier classifiers, while the backbone network is associated with comparatively low gradient norms. This restricts the model's ability to effectively capture and adapt to features from newly introduced camera domains. In contrast, the proposed ETF strategy, owing to its intrinsic geometric properties, provides more informative and discriminative supervisory signals. The observed increase in gradient norms indicates that the model undergoes more substantial parameter updates, thereby better activating both shallow and deep feature representations. Such a more effective optimization dynamic not only facilitates

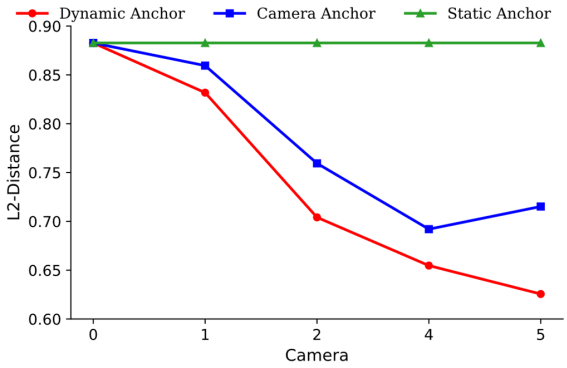


Fig. 13. By comparing the distances between different prototype mechanisms and feature anchors, the representational fidelity of the prototypes can be evaluated. The smaller the distance, the stronger the fidelity.

TABLE V
IDEAL REFERENCE ANGLES FOR EXACT K-CLASS SIMPLEX-ETFS.

Number of classes K	Ideal simplex-ETF angle
10	96.379°
50	91.169°
100	90.579°
751 (Market-1501)	90.076°
1041 (MSMT17)	90.055°
10000	90.006°

faster convergence but also enhances the model's adaptability to cross-camera variations.

Geometric Stability of Dynamic Target Expansion. To evaluate the stability of the fixed-target geometry, we compute the pairwise angle between every two normalized targets:

$$\theta_{ij}^{t} = \arccos\left(\left\langle w_i^t, w_j^t \right\rangle\right), \qquad i \neq j.$$

The mean angle describes the global angular scale, whereas the minimum angle measures the worst-case local separation and is therefore more sensitive to geometric crowding.

For an ideal K-class simplex-ETF, every pair satisfies

$$\langle w_i, w_j \rangle = -\frac{1}{K-1}, \qquad \theta_{\mathrm{ETF}}(K) = \arccos\left(-\frac{1}{K-1}\right).$$

As shown in Table V, although this reference angle depends on K, it rapidly approaches $90°$ as the class number increases.

When an orthogonal complement is available, DGA appends a target orthogonal to the established target span, producing a $90°$ relation to the old targets. This is not an exact finite-K simplex-ETF, but an orthogonalized large-class approximation. We quantify its deviation as

$$\delta_{\mathrm{ETF}}(t) = \max_{i \neq j} \left| \left\langle w_i^t, w_j^t \right\rangle + \frac{1}{K_t - 1} \right|.$$

As shown in Fig. 11, the mean angle remains close to $90°$, consistent with both the orthogonalized construction and the large-class simplex-ETF reference. More importantly, the minimum angle remains near $85°$ and does not decrease continuously as identities are added, indicating that the worst-separated target pair does not undergo progressive crowding or collapse. These results support the stability of the expanded

target geometry without implying that DGA constructs an exact finite-K simplex-ETF.

Analysis of Fragmented Feature Aggregation. We compare three prototype mechanisms for identity y. The camera-specific anchor uses only the observations from camera stage t:

$$C_y^t = \mathrm{Norm}\left(\frac{1}{N_y^t}\sum_{i:y_i=y} f_{\theta_t}(x_i^t)\right).$$

The static anchor is initialized when the identity is first observed at stage t_0 and is then frozen:

$$S_y = C_y^{t_0}.$$

The proposed dynamic global prototype accumulates observations through normalized momentum consolidation:

$$P_y^t = \mathrm{Norm}\left(mP_y^{t-1} + (1-m)C_y^t\right).$$

For post-hoc evaluation only, we compute an offline global identity center using all available cross-camera observations. This center is never accessed during incremental training. Fig. 13 reports the L2 distance from each prototype mechanism to this offline reference. All three anchors coincide when only the initial camera is available. The static anchor then remains near 0.88 because it never absorbs later observations. The camera-specific anchor becomes closer to the global center from C1 to C4, but its distance increases from approximately 0.69 at C4 to 0.72 at C5. This rebound reflects its dependence on the most recent local view and is consistent with the larger domain discrepancy of C5. In contrast, the dynamic global prototype decreases steadily from approximately 0.88 at C0 to 0.63 at C5, including a further improvement when C5 is introduced. By retaining historical evidence while incorporating the new observation, normalized prototype consolidation reduces sensitivity to the latest camera shift and provides a more faithful approximation to the global identity distribution than either a single-camera or frozen anchor.

Computational and Storage Cost. We conducted a performance analysis comparing the baseline model and CIGRE under identical conditions regarding hardware, batch size, input resolution, camera sequence, and training configuration. For the Market-1501 dataset, the average runtime per run was calculated based on the mean GPU time over 60 epochs; CCIC and DGA were also measured as part of the internal analysis.

Compared with the baseline, complete CIGRE adds 0.188 s of GPU time per epoch, corresponding to 1.83%, and increases peak allocated memory by 12.973 MB, corresponding to 0.18%. DGA is triggered primarily when the identity set expands, while CCIC uses compact vector operations and introduces no additional backbone branch. At inference, the target and prototype memories are discarded, and the deployed model has the same architecture, parameter count, and FLOPs as the baseline.

TABLE VI
COMPUTATIONAL OVERHEAD OF DIFFERENT FRAMEWORK COMPONENTS.

Method	GPU time (s)	GPU memory (MB)
Baseline	10.301	7345.576
CIGRE	10.489	7358.549
Baseline + CCIC	10.405	7348.781
Baseline + DGA	10.322	7347.836

V. CONCLUSION

In this paper, we investigate the Camera-Incremental Person Re-Identification (CI-ReID) problem, which models the continual expansion of camera networks under identity overlap, domain shifts, and replay-free constraints. To address feature-space distortion and representation fragmentation, we propose Camera-Incremental Geometric Representation Evolution (CIGRE). Specifically, Dynamic Geometric Alignment (DGA) maintains a stable and expandable global discriminative structure, while Cross-Camera Identity Consolidation (CCIC) progressively integrates fragmented identity representations across newly introduced camera views. Extensive experiments on Market-1501 and MSMT17 demonstrate that CIGRE consistently improves seen-camera retrieval and unseen-camera generalization.

In our future work, we will explore a wider range of experimental setups and privacy protection issues. The primary protocol still relies on globally consistent identity annotations, and extending CIGRE to weakly supervised or unsupervised cross-stage association is an important direction. In addition, replay-free prototype storage does not provide a formal privacy guarantee, motivating future study of privacy-preserving prototype learning.

REFERENCES

[1] Z. Lu, W. Shi, H. Li, and M. Ye, "Prototype-guided knowledge propagation with adaptive learning for lifelong person re-identification," in *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence*, 2025, pp. 5851–5859.

[2] M. Ye, S. Chen, C. Li, W.-S. Zheng, D. Crandall, and B. Du, "Transformer for object re-identification: A survey," *International Journal of Computer Vision*, pp. 2410–2440, 2025.

[3] S. Chen, M. Ye, and B. Du, "Rotation invariant transformer for recognizing object in uavs," in *Proceedings of the 30th ACM International Conference on Multimedia*, 2022, pp. 2565–2574.

[4] M. Ye, J. Shen, G. Lin, T. Xiang, L. Shao, and S. C. H. Hoi, "Deep learning for person re-identification: A survey and outlook," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, pp. 2872–2893, 2022.

[5] Q. Meng, H. Li, M. Cao, and M. Ye, "Towards real-world holistic privacy-preserving person re-identification," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, pp. 1–13, 2026.

[6] G. Wu and S. Gong, "Generalising without forgetting for lifelong person re-identification," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2021, pp. 2889–2897.

[7] J. Liu, Z.-J. Zha, D. Chen, R. Hong, and M. Wang, "Adaptive transfer network for cross-domain person re-identification," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 7202–7211.

[8] D. Goswami, Y. Liu, B. Twardowski, and J. Van De Weijer, "Fecam: Exploiting the heterogeneity of class distributions in exemplar-free continual learning," *Advances in Neural Information Processing Systems*, pp. 6582–6595, 2023.

[9] J. Huang, X. Yu, D. An, Y. Wei, X. Bai, J. Zheng, C. Wang, and J. Zhou, "Learning consistent region features for lifelong person re-identification," *Pattern Recognition*, p. 109837, 2023.

[10] Z. Mai, R. Li, J. Jeong, D. Quispe, H. Kim, and S. Sanner, "Online continual learning in image classification: An empirical survey," *Neurocomputing*, pp. 28–51, 2022.

[11] R. Aljundi, E. Belilovsky, T. Tuytelaars, L. Charlin, M. Caccia, M. Lin, and L. Page-Caccia, "Online continual learning with maximal interfered retrieval," *Advances in Neural Information Processing Systems*, 2019.

[12] W. Shi and M. Ye, "Prototype reminiscence and augmented asymmetric knowledge aggregation for non-exemplar class-incremental learning," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 1772–1781.

[13] A. Yadav and D. K. Vishwakarma, "Deep learning algorithms for person re-identification: sate-of-the-art and research challenges," *Multimedia Tools and Applications*, pp. 22 005–22 054, 2024.

[14] Z. Zhong, L. Zheng, Z. Zheng, S. Li, and Y. Yang, "Camera style adaptation for person re-identification," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 5157–5166.

[15] Z. Sun and Y. MU, "Patch-based knowledge distillation for lifelong person re-identification," in *Proceedings of the 30th ACM International Conference on Multimedia*, 2022, p. 696–707.

[16] L. Zhang, G. Gao, and H. Zhang, "Spatial–temporal federated learning for lifelong person re-identification on distributed edges," *IEEE Transactions on Circuits and Systems for Video Technology*, pp. 1884–1896, 2023.

[17] C. Yu, Y. Shi, Z. Liu, S. Gao, and J. Wang, "Lifelong person re-identification via knowledge refreshing and consolidation," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2023, pp. 3295–3303.

[18] W. Ge, J. Du, A. Wu, Y. Xian, K. Yan, F. Huang, and W.-S. Zheng, "Lifelong person re-identification by pseudo task knowledge preservation," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2022, pp. 688–696.

[19] N. Pu, W. Chen, Y. Liu, E. M. Bakker, and M. S. Lew, "Lifelong person re-identification via adaptive knowledge accumulation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 7901–7910.

[20] N. Asadi, M. Davari, S. Mudur, R. Aljundi, and E. Belilovsky, "Prototype-sample relation distillation: towards replay-free continual learning," in *International Conference on Machine Learning*, 2023, pp. 1093–1106.

[21] S. He, H. Luo, P. Wang, F. Wang, H. Li, and W. Jiang, "Transreid: Transformer-based object re-identification," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 15 013–15 022.

[22] C. Wang, Y. Yang, M. Qi, H. Zhang, and H. Ma, "Towards efficient object re-identification with a novel cloud-edge collaborative framework," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025, pp. 7600–7608.

[23] X. Gong, X. Tan, and Y. Xiang, "Contrastive mean teacher for intra-camera supervised person re-identification," *IEEE Transactions on Circuits and Systems for Video Technology*, pp. 9786–9797, 2024.

[24] Y.-C. Chen, X. Zhu, W.-S. Zheng, and J.-H. Lai, "Person re-identification by camera correlation aware feature augmentation," *IEEE transactions on pattern analysis and machine intelligence*, pp. 392–408, 2017.

[25] X. Wang, Q. Zhu, S. Wu, B. Jiang, and S. Zhang, "When person re-identification meets event camera: a benchmark dataset and an attribute-guided re-identification framework," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026, pp. 10 172–10 180.

[26] Y. Yang, S. Chen, X. Li, L. Xie, Z. Lin, and D. Tao, "Inducing neural collapse in imbalanced learning: Do we really need a learnable classifier at the end of deep neural network?" *Advances in Neural Information Processing Systems*, pp. 37 991–38 002, 2022.

[27] E. Markou, T. Ajanthan, and S. Gould, "Guiding neural collapse: Optimising towards the nearest simplex equiangular tight frame," *Advances in Neural Information Processing Systems*, pp. 35 544–35 573, 2024.

[28] M. Seo, H. Koh, W. Jeung, M. Lee, S. Kim, H. Lee, S. Cho, S. Choi, H. Kim, and J. Choi, "Learning equi-angular representations for online continual learning," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 23 933–23 942.

[29] D.-W. Zhou, Q.-W. Wang, Z.-H. Qi, H.-J. Ye, D.-C. Zhan, and Z. Liu, "Class-incremental learning: A survey," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, pp. 9851–9873, 2024.

[30] D.-W. Zhou, Z.-W. Cai, H.-J. Ye, D.-C. Zhan, and Z. Liu, "Revisiting class-incremental learning with pre-trained models: Generalizability and adaptivity are all you need," *International Journal of Computer Vision*, pp. 1012–1032, 2025.

[31] G. M. Van de Ven, T. Tuytelaars, and A. S. Tolias, "Three types of incremental learning," *Nature Machine Intelligence*, pp. 1185–1197, 2022.

[32] W. Shi and M. Ye, "Prospective representation learning for non-exemplar class-incremental learning," *Advances in Neural Information Processing Systems*, pp. 995–1018, 2024.

[33] D.-W. Zhou, Z.-W. Cai, H.-J. Ye, L. Zhang, and D.-C. Zhan, "Dual consolidation for pre-trained model-based domain-incremental learning," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 20 547–20 557.

[34] M. Masana, X. Liu, B. Twardowski, M. Menta, A. D. Bagdanov, and J. Van De Weijer, "Class-incremental learning: survey and performance evaluation on image classification," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, pp. 5513–5533, 2022.

[35] K. Xu, X. Zou, G. Hua, and J. Zhou, "Componential prompt-knowledge alignment for domain incremental learning," in *International Conference on Machine Learning*, 2025, pp. 70 032–70 046.

[36] L. Zheng, H. Zhang, S. Sun, M. Chandraker, Y. Yang, and Q. Tian, "Person re-identification in the wild," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017, pp. 1367–1376.

[37] Y. Zou, X. Yang, Z. Yu, B. V. Kumar, and J. Kautz, "Joint disentangling and adaptation for cross-domain person re-identification," in *European Conference on Computer Vision*, 2020, pp. 87–104.

[38] Q. Yang, A. Wu, and W.-S. Zheng, "Person re-identification by contour sketch under moderate clothing change," *IEEE transactions on pattern analysis and machine intelligence*, pp. 2029–2046, 2019.

[39] Q. Li, K. Xu, Y. Peng, and J. Zhou, "Exemplar-free lifelong person re-identification via prompt-guided adaptive knowledge consolidation," *International Journal of Computer Vision*, pp. 4850–4865, 2024.

[40] Z. Huang, Z. Zhang, C. Lan, W. Zeng, P. Chu, Q. You, J. Wang, Z. Liu, and Z.-J. Zha, "Lifelong unsupervised domain adaptive person re-identification with coordinated anti-forgetting and adaptation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 14 288–14 297.

[41] Z. Cui, J. Zhou, and Y. Peng, "Bi-c 2 r: Bidirectional continual compatible representation for re-indexing free lifelong person re-identification," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025.

[42] L. Wei, S. Zhang, W. Gao, and Q. Tian, "Person transfer gan to bridge domain gap for person re-identification," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2018, pp. 79–88.

[43] Z. Yang, D. Wu, C. Wu, Z. Lin, J. Gu, and W. Wang, "A pedestrian is worth one prompt: Towards language guidance person re-identification," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 17 343–17 353.

[44] Y. Wei, J. Ye, Z. Huang, J. Zhang, and H. Shan, "Online prototype learning for online continual learning," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 18 764–18 774.

[45] T. Yu, D. Li, Y. Yang, T. M. Hospedales, and T. Xiang, "Robust person re-identification by modelling feature uncertainty," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 552–561.

[46] Z. Yang, D. Wu, W. Zhang, B. Li, and W. Wang, "Handling label uncertainty for camera incremental person re-identification," in *Proceedings of the 31st ACM International Conference on Multimedia*, 2023, pp. 6253–6263.

[47] H. Yao, J. Luo, L. Yu, and C. Xu, "Camera-incremental object re-identification with identity knowledge evolution," *IEEE Transactions on Multimedia*, pp. 11 246–11 260, 2024.

[48] Y.-X. Peng, Y. Li, and W.-S. Zheng, "Revisiting person re-identification by camera selection," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, pp. 2692–2708, 2023.

[49] N. Pu, Z. Zhong, N. Sebe, and M. S. Lew, "A memorizing and generalizing framework for lifelong person re-identification," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, pp. 13 567–13 585, 2023.

[50] Z. Yang, D. Wu, W. Zhang, J. Gu, Z. Lin, and W. Wang, "Privacy-preserving replay and adaptive relation distillation for camera incremental person re-identification," in *IEEE International Conference on Multimedia and Expo*, 2024, pp. 1–6.

[51] X. Zhu, X. Zhu, M. Li, P. Morerio, V. Murino, and S. Gong, "Intra-camera supervised person re-identification," *International Journal of Computer Vision*, pp. 1580–1595, 2021.

[52] K. Xu, H. Cheng, J. Li, X. Zou, and J. Zhou, "Vision-language attribute disentanglement and reinforcement for lifelong person re-identification," *arXiv preprint arXiv:2603.19678*, 2026.

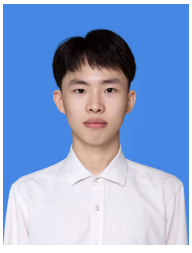
[53] H. Chen, F. Bremond, N. Sebe, and S. Zhang, "Anti-forgetting adaptation for unsupervised person re-identification," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, pp. 1056–1072, 2024.

[54] Z. Cui, J. Zhou, X. Wang, M. Zhu, and Y. Peng, "Learning continual compatible representation for re-indexing free lifelong person re-identification," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 16 614–16 623.

[55] Y. Lu, M. Wang, and W. Deng, "Augmented geometric distillation for data-free incremental person reid," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 7329–7338.
[56] N. Pu, Y. Liu, W. Chen, E. M. Bakker, and M. S. Lew, "Meta reconciliation normalization for lifelong person re-identification," in *Proceedings of the 30th ACM International Conference on Multimedia*, 2022, pp. 541–549.
[57] X. Hao, S. Zhao, M. Ye, and J. Shen, "Cross-modality person re-identification via modality confusion and center aggregation," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 16 403–16 412.
[58] K. Xu, C. Jiang, P. Xiong, Y. Peng, and J. Zhou, "Dask: Distribution rehearsing via adaptive style kernel learning for exemplar-free lifelong person re-identification," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025, pp. 8915–8923.
[59] K. Xu, Z. Liu, X. Zou, Y. Peng, and J. Zhou, "Long short-term knowledge decomposition and consolidation for lifelong person re-identification," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025.
[60] J. Zhou, K. Xu, F. Zhuo, X. Zou, and Y. Peng, "Distribution-aware knowledge aligning and prototyping for non-exemplar lifelong person re-identification," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, pp. 10 932–10 948, 2025.
[61] C. Liu, X. Chang, and Y.-D. Shen, "Unity style transfer for person re-identification," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 6887–6896.
[62] Y. Chong, C. Peng, J. Zhang, and S. Pan, "Style transfer for unsupervised domain-adaptive person re-identification," *Neurocomputing*, pp. 314–321, 2021.
[63] J. Wu, H. Liu, W. Shi, M. Liu, and W. Li, "Style-agnostic representation learning for visible-infrared person re-identification," *IEEE Transactions on Multimedia*, pp. 2263–2275, 2023.
[64] V. Papyan, X. Y. Han, and D. L. Donoho, "Prevalence of neural collapse during the terminal phase of deep learning training," *Proceedings of the National Academy of Sciences*, pp. 24 652–24 663, 2020.
[65] V. Kothapalli, "Neural collapse: A review on modelling principles and generalization," *arXiv preprint arXiv:2206.04041*, 2022.
[66] Y. Wang, J. Sun, C. Wang, M. Zhang, and M. Yang, "Navigate beyond shortcuts: Debiased learning through the lens of neural collapse," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 12 322–12 331.
[67] E. Markou, T. Ajanthan, and S. Gould, "Guiding neural collapse: Optimising towards the nearest simplex equiangular tight frame," *Advances in Neural Information Processing Systems*, pp. 35 544–35 573, 2024.
[68] Z. Zhu, T. Ding, J. Zhou, X. Li, C. You, J. Sulam, and Q. Qu, "A geometric analysis of neural collapse with unconstrained features," *Advances in Neural Information Processing Systems*, pp. 29 820–29 834, 2021.
[69] Y. Yang, S. Chen, X. Li, L. Xie, Z. Lin, and D. Tao, "Inducing neural collapse in imbalanced learning: Do we really need a learnable classifier at the end of deep neural network?" *Advances in Neural Information Processing Systems*, pp. 37 991–38 002, 2022.
[70] S.-A. Rebuffi, A. Kolesnikov, G. Sperl, and C. H. Lampert, "icarl: Incremental classifier and representation learning," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017, pp. 2001–2010.
[71] V. Kothapalli, "Neural collapse: A review on modelling principles and generalization," *Transactions on Machine Learning Research*, 2023.
[72] L. Zheng, L. Shen, L. Tian, S. Wang, J. Wang, and Q. Tian, "Scalable person re-identification: A benchmark," in *Proceedings of the IEEE International Conference on Computer Vision*, 2015, pp. 1116–1124.
[73] J. Shi, Y. Zhang, X. Yin, Y. Xie, Z. Zhang, J. Fan, Z. Shi, and Y. Qu, "Dual pseudo-labels interactive self-training for semi-supervised visible-infrared person re-identification," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 11 218–11 228.
[74] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 770–778.
[75] Z. Cui, J. Zhou, and Y. Peng, "Dkc: Differentiated knowledge consolidation for cloth-hybrid lifelong person re-identification," in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 3573–3582.
[76] K. Xu, X. Zou, Y. Peng, and J. Zhou, "Distribution-aware knowledge prototyping for non-exemplar lifelong person re-identification," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 16 604–16 613.
[77] K. Xu, X. Zou, and J. Zhou, "Lstkc: Long short-term knowledge consolidation for lifelong person re-identification," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2024, pp. 16 202–16 210.
[78] K. Xu, Z. Liu, X. Zou, Y. Peng, and J. Zhou, "Long short-term knowledge decomposition and consolidation for lifelong person re-identification," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2025.
[79] Z. Li and D. Hoiem, "Learning without forgetting," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, pp. 2935–2947, 2017.
[80] F. Tung and G. Mori, "Similarity-preserving knowledge distillation," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 1365–1374.
[81] N. Asadi, M. Davari, S. Mudur, R. Aljundi, and E. Belilovsky, "Prototype-sample relation distillation: towards replay-free continual learning," in *International Conference on Machine Learning*, 2023, pp. 1093–1106.
[82] S. He, H. Luo, P. Wang, F. Wang, H. Li, and W. Jiang, "Transreid: Transformer-based object re-identification," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 15 013–15 022.
[83] D. Fu, D. Chen, J. Bao, H. Yang, L. Yuan, L. Zhang, H. Li, and D. Chen, "Unsupervised pre-training for person re-identification," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2021, pp. 14 750–14 759.
[84] D. Fu, D. Chen, H. Yang, J. Bao, L. Yuan, L. Zhang, H. Li, F. Wen, and D. Chen, "Large-scale pre-training for person re-identification with noisy labels," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 2476–2486.
[85] K. Zhu, H. Guo, T. Yan, Y. Zhu, J. Wang, and M. Tang, "Pass: Part-aware self-supervised pre-training for person re-identification," in *European Conference on Computer Vision*, 2022, pp. 198–214.
[86] J. Zuo, Y. Nie, H. Zhou, H. Zhang, H. Wang, T. Guo, N. Sang, and C. Gao, "Cross-video identity correlating for person re-identification pre-training," *Advances in Neural Information Processing Systems*, pp. 25 228–25 250, 2024.



Mang Ye (Senior Member, IEEE) received the Ph.D. degree in computer science from Hong Kong Baptist University in 2019. He is currently a Full Professor with the School of Computer Science, Wuhan University, Wuhan, China. He was a Research Scientist at Inception Institute of Artificial Intelligence, Abu Dhabi, UAE. He is currently a Full professor at Wuhan University. He serves as an Associate Editor for IEEE Transactions on Information Forensics and Security and IEEE Transactions on Image Processing. He also serve as an Area Chair for ICML, ICLR, CVPR, ECCV, ACM MM etc. His research interests include computer vision, pattern recognition, and federated learning.



Shipeng Zhang is currently pursuing the B.E. degree in Software Engineering at the School of Computer Science, Wuhan University. His research interests include Continual Learning and Person Re-Identification.



Wuxuan Shi is currently a Ph.D. candidate with the School of Computer Science, Wuhan University. He received his B.S. degree from Wuhan University in 2022. His current research interests include Computer Vision and Continual Learning.



Zhijie Lu is currently pursuing the B.E. degree in Computer Science at the School of Computer Science, Wuhan University, China. His research interests include Person Re-Identification and Continual Learning.



Xingyu Gao received his Ph.D. in Computer Science from the Institute of Computing Technology, Chinese Academy of Sciences. He was a visiting scholar at Nanyang University and Singapore Management University. Currently, he is a researcher at the Institute of Microelectronics, Chinese Academy of Sciences. His research interests include Machine Learning, Multimedia Content Analysis and Search, and Computer Vision.



He Li received the B.A. mod and M.Sc. de-grees in Computer Science from Trinity CollegeDublin, Ireland, in 2019 and 2021. He receivedthe Ph.D. degree in Artificial Intelligence from Wuhan University. He is currently a PostDoc. at the School of Computer Science, Wuhan Uni-versity, China. His research interests focus on Person Re-Identification and Al for Medicine.